SAIT – ARTI 406 – Human Centered AI

Final Project Report

AI Grading System

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Introduction

Grading remains one of the most persistent bottlenecks in education, since teachers and instructors must devote a substantial amount of time to reviewing assignments, applying rubrics consistently, and writing meaningful feedback, all of which reduces the time they can spend on direct instruction or student support. As class sizes increase, this challenge becomes even more difficult to manage, because maintaining both efficiency and consistency across large numbers of submissions places a significant strain on instructors' daily workload.

To address this problem, this project introduces the AI Grading System, a web-based platform designed to assist instructors by reducing the processing burden associated with evaluating exams, quizzes, projects, and other academic submissions. The system allows users to upload work in standard formats such as PDF, DOCX, and TXT, as well as less structured formats including images, presentations, and recordings, after which artificial intelligence analyzes the submission in relation to a specified grading rubric.

The purpose of the platform is not to replace the instructor, but to provide a flexible tool that improves efficiency while preserving academic oversight, especially through features that allow users to adjust grading rigor to strict, medium, or easy depending on their instructional goals. After processing a submission, the system generates a detailed evaluation that includes a letter grade, a percentage score, and written feedback describing the student's strengths and areas for improvement, all of which can then be reviewed and manually adjusted by the instructor before any grade is finalized.

Analyze User Needs

Designing an effective educational tool required more than identifying a useful application of AI, since it was equally important to understand how instructors would actually move through the system, interpret its interface, and respond to its workflow during real use. During the early development phase, we therefore focused heavily on the user interface and conducted practical feedback sessions with three instructors using UI prototypes, and because no active AI model had yet been integrated, the evaluation centered entirely on workflow clarity, layout decisions, and overall usability rather than model performance itself.

Although the instructors responded positively to the general concept and immediately recognized the potential value of a system that could assist with grading, their interactions with the mockups made it clear that the initial design introduced unnecessary friction in several areas. Uzma, for example, questioned why the interface asked the instructor to manually enter their name when that information should instead be tied to the account used to log in, and she also pointed out that fields populated automatically from uploaded documents should not resemble editable input fields, since a read-only display would more clearly communicate that the information had already been extracted by the system. She further noted that the earlier workflow, which allowed multiple student documents to be uploaded at once, made the process feel more complicated than necessary, and she preferred a simpler approach in which one student submission would be graded at a time with a clear option to add the next document afterward.

Reeta's feedback reinforced many of the same concerns, especially with respect to clarity and simplicity, which suggested that these were not isolated observations but recurring usability problems in the original design. In addition to echoing Uzma's comments, she specifically identified confusion within the grade overview section at the end of the workflow, where the available export methods, particularly the email option, were not immediately intuitive and therefore created uncertainty about how final results were meant to be saved or shared.

Steven's feedback shifted the discussion from interface clarity alone to the broader issue of trust in AI-assisted grading, particularly in situations where the model's interpretation may not fully match the instructor's intent. While he responded positively to the overall mockup, he emphasized that instructors would need access to a clear explanation of how grades were generated, along with a way to correct or override results in cases where the AI misread the rubric, and he also noted that AI systems often tend toward overly positive feedback, which made it important to include a mechanism for steering the system toward a stricter or more appropriate interpretation. In response to this concern, the system was updated to include an edit and override option for grades, strengthening instructor control and ensuring that AI output remains subject to human review rather than being treated as final.

Taken together, this feedback demonstrated that the most important design priority was not necessarily adding more features, but reducing ambiguity and making the workflow easier to understand from one step to the next. By simplifying how information is presented, clarifying which fields are auto-filled, shifting the process toward single-submission grading, and

improving instructor control over final results, the revised interface became more intuitive and better aligned with the practical expectations of its intended users.

Research Questions

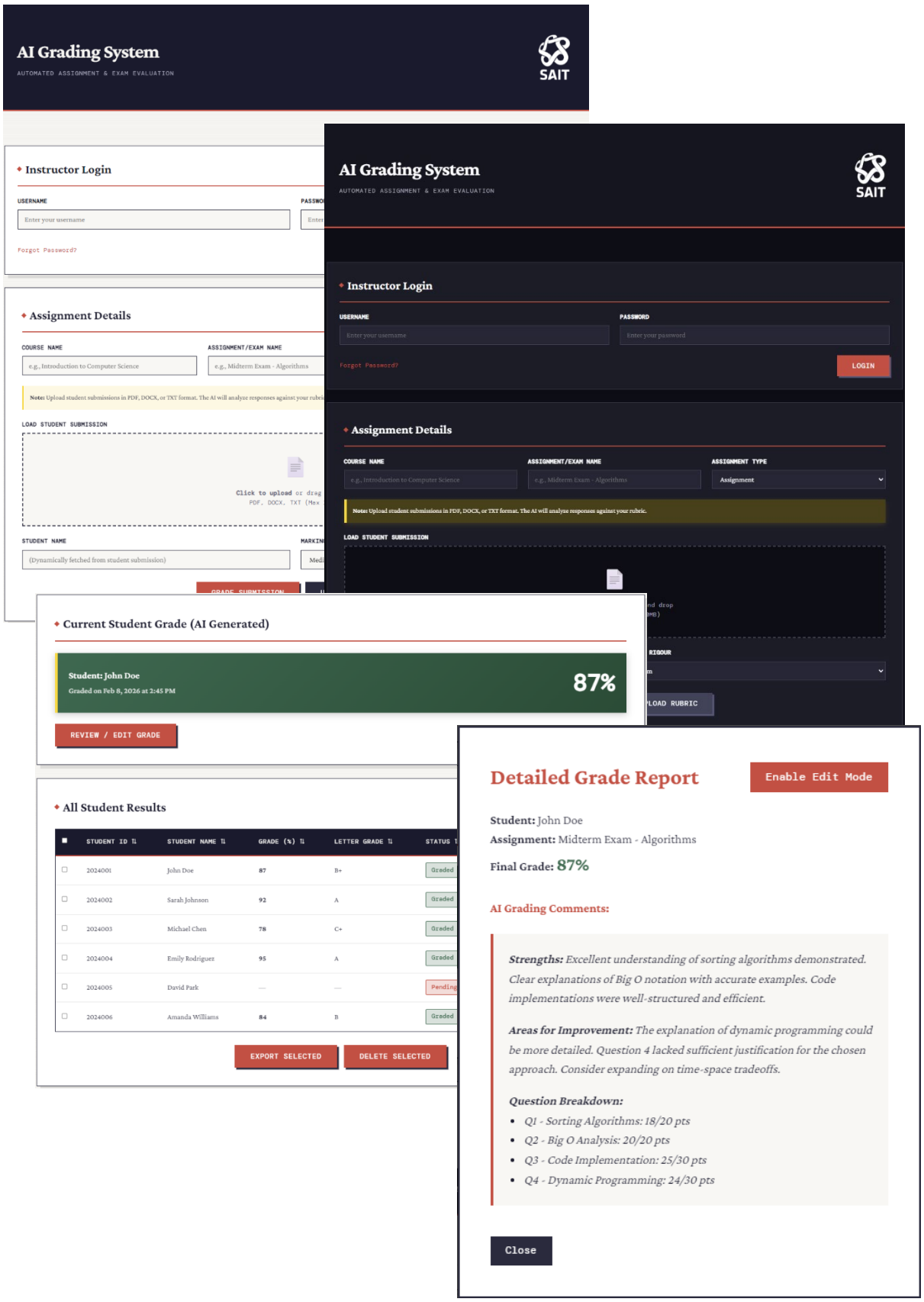
Because this project focused on the ideation and design of an AI grading application rather than the implementation of a working model, the purpose of this stage was to define the key questions that would guide future development if the system were to be built. The project therefore centered on identifying what the AI would need to do, what risks would need to be addressed, and what conditions would have to be met for such a system to be useful and responsible in an educational setting.

The first research question concerned comprehension, specifically whether an AI model would be capable of accurately interpreting an uploaded grading rubric and applying that rubric consistently across different types of student submissions. Since the proposed system allows instructors to upload their own rubrics and adjust the marking rigor to strict, medium, or easy, the effectiveness of the application would depend on whether the model could follow detailed grading instructions and adapt to the expectations of different educators without losing consistency.

The second research question focused on transparency, since an AI-generated grade would only be useful if the instructor could understand how that result was produced and judge whether the reasoning was sound. In the proposed workflow, the instructor remains responsible for the final mark, so the system would need to generate more than a score alone by also providing a clear explanation of the student's strengths, weaknesses, and overall performance in relation to the rubric.

The final research question addressed privacy and ethics, as any system designed to process student coursework would also need to handle sensitive academic information in a secure and fair manner. This raised important questions about how student data would be stored, how it would be protected during analysis, and whether the proposed grading process could avoid bias while still producing evaluations that instructors could trust.

User Interface Design



ML/AI Model

Because the AI grading system was not implemented as part of this project, the model design presented here describes a proposed architecture rather than a completed backend. The goal of this section is to outline how such a system could operate in practice if development were to continue beyond the design stage.

In the proposed architecture, the grading pipeline would begin with document processing tools capable of extracting text from uploaded submissions, including standard files such as PDFs and word-processing documents, as well as less structured sources that might require optical character recognition. Once the submission had been converted into usable text, that content would be passed to a natural language processing model, most likely a large language model, which would be responsible for comparing the student's work against the instructor's uploaded rubric.

Rather than treating grading as a single step, the proposed model would evaluate a submission by breaking the task into smaller decisions aligned with the rubric criteria. In this design, the AI would first identify which portions of the student's response correspond to specific grading categories and would then classify the quality of those responses according to rubric-defined performance levels such as poor, good, or excellent. This approach was selected because it offers a more structured and interpretable path to grading than generating a single overall score without explanation.

The model would also need to account for instructor-selected grading rigor, meaning that the system would not simply apply one fixed standard to every submission regardless of context. Instead, the proposed application would translate settings such as strict, medium, or easy into instructions that slightly adjust how the model interprets quality thresholds, after which it would combine its criterion-level judgments into a final percentage, a letter grade, and written feedback describing the student's strengths and areas for improvement.

Evaluation/Testing Process

The first stage of evaluation would involve comparing AI-generated grades against previously marked student submissions that had already been assessed by instructors. This would make it possible to examine how closely the model's scores aligned with human judgment, while also testing whether the system could apply rubric categories consistently across a range of

assignment types and response qualities. Because the proposed backend relies on criterion-level classification, relevant metrics could include agreement with instructor grading, accuracy across rubric categories, and measures such as precision and recall where appropriate.

A second stage of testing would focus on system robustness by examining how well the grading pipeline handles different document types, formatting irregularities, and variations in submission quality. This would be especially important for verifying that text extraction remains reliable when students upload poorly formatted PDFs, simple text files, or other materials that do not follow a perfectly standardized structure. The rigor settings would also need to be tested by running the same submission through multiple grading modes to determine whether the output changes in a controlled and interpretable way when the instructor selects strict, medium, or easy evaluation criteria.

The final stage of evaluation would involve user acceptance testing with instructors, since technical accuracy alone would not be enough to establish trust in an educational environment. In this stage, instructors would review the AI-generated grades and feedback through the dashboard, judge whether the explanations are reasonable, and determine whether the system's outputs align with their own professional expectations. If instructors repeatedly felt the need to override grades or correct misleading feedback, those patterns would provide valuable evidence about where the model or prompt design would need refinement.

Ethical Considerations

Ethical concerns are central to the design of this project because the proposed application would process student work that may contain personally identifiable and academically sensitive information. Even at the concept stage, it is necessary to consider how such a system could protect privacy, reduce unfair bias, and preserve meaningful human oversight in the grading process.

One of the most immediate concerns involves data security, particularly if student submissions, names, and identification numbers are processed through a third-party AI provider. Any future implementation would need to ensure that student documents are not retained or reused for external model training without authorization, and this raises an important design decision between relying on an outside service and building a self-hosted institutional solution. While a locally hosted model would require greater technical effort, it would offer stronger protection by keeping student data within the institution's own systems.

Bias is another major ethical issue, since both custom-trained and off-the-shelf models can reproduce unfair patterns in the way they interpret language, structure feedback, or reward particular writing styles. A grading system that favors certain forms of expression over others could disadvantage students unfairly, especially those whose language background or communication style differs from the patterns most common in the training data. For that reason, any future implementation would need to be evaluated not only for grading consistency, but also for whether it assesses the substance of a student's work rather than superficial features of phrasing or tone.

Human oversight is therefore essential to the ethical use of a system like this, because automated grading should function as a support tool rather than a final authority. The proposed design addresses this by keeping the instructor in control of the final grade, providing editable outputs, and requiring the system to present understandable feedback so that its reasoning can be reviewed rather than blindly accepted. In this way, the project frames AI as a mechanism for reducing workload and improving efficiency while still preserving the educator's responsibility for academic judgment.

Conclusion

This project presented the design and conceptual development of an AI Grading System intended to reduce the time instructors spend evaluating assignments while preserving their control over the final grading process. By focusing on user needs, interface clarity, and responsible AI integration, the report outlined a practical application that could support educators by generating rubric-based grades, percentage scores, and written feedback in a more efficient and structured workflow.

A central theme throughout the project was that automation in education should function as assistance rather than replacement. For that reason, the proposed system was designed to keep instructors in the loop through adjustable grading rigor, transparent feedback, and the ability to review, edit, or override AI-generated results before any grade is finalized. The instructor feedback gathered during the design stage further reinforced that usability, clarity, and trust would be essential to the success of any future implementation.

Although the AI model itself was not built as part of this project, the report established a clear foundation for what such a system would require if developed further, including document processing, rubric interpretation, structured evaluation logic, and a testing framework centered

on both technical performance and instructor acceptance. At the same time, the project recognized that any real deployment would need to address serious concerns related to privacy, bias, and data security before it could be used in an academic setting.

Overall, this project demonstrates that an AI-assisted grading platform is a realistic and potentially valuable direction for educational technology, provided that efficiency is balanced with transparency, fairness, and human oversight. Rather than presenting AI as a substitute for professional judgment, the proposed system positions it as a tool that could help instructors reclaim time for the parts of teaching that require the most human attention, including guidance, mentorship, and meaningful engagement with students.